

Impact of Prompt Engineering Training on Learning Outcomes in an Engineering Course

Section A

No Training

n = 15 students

Section B

Moderate

n = 15 students

Section C

Extended

n = 15 students

Same instructor Same lectures 4 Assignments (each requires Chat-GPT; logs submitted) A1: context & step-by-step A2: expectations & multimodal A3: decomposition & criteria A4: roles & prompt comparison

Pre & Post Survey

9 dimensions (UTAUT2):
trust, usefulness,
ease of use,
locus of control, ...

Grades

Midterm & Final grades
(same scale
across sections)

Prompt Quality

Instructor-rated 1–5
(chat logs reviewed)

Analysis: One-way ANOVA (section differences) Pearson correlations Thematic analysis of open-ended responses (grounded theory)

Key Finding 1

Extended training **significantly** improves prompt quality ($p = 0.0017$; mean A2.91 to C3.57)

Key Finding 2

Better prompts correlate with higher final grades ($r = 0.45$, $p < 0.001$)

Key Finding 3

Higher perceived control to lower prompt quality (locus of control: $r = -0.57$)